

Introduction. What is meta-analysis?

G0B75A Meta-analysis

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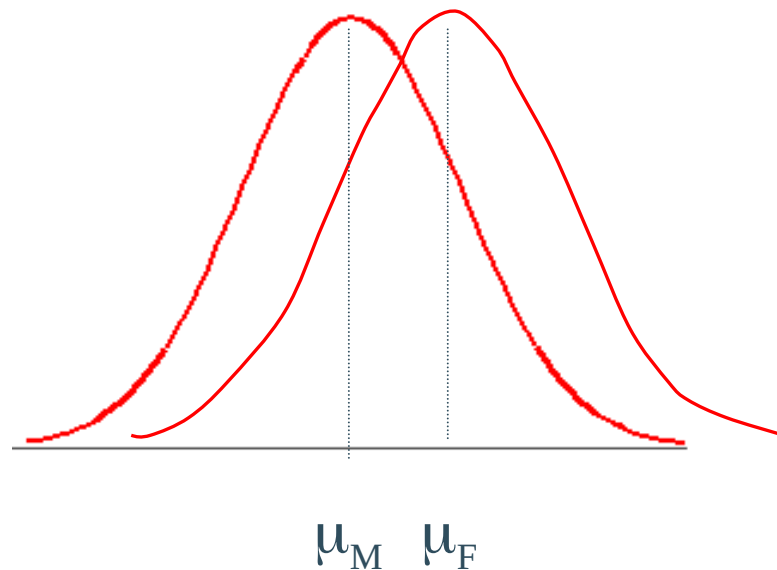
Outline

- Introduction: Failures to replicate. A simulation.
- More complex meta-analyses
- Meta-analytic questions and steps

Failures to replicate

Example: association between gender and math

$$\mu_M = 8 ; \quad \mu_F = 8.5 ; \quad \sigma_M = \sigma_F = 1.5$$



$$\delta = \frac{\mu_F - \mu_M}{\sigma} = 0.33$$

Standardized mean difference (Cohen, 1969):

$$\delta = \frac{\mu_A - \mu_B}{\sigma} \quad \text{'True' effect size}$$

Estimated by its sample counterpart:

$$g = \frac{\bar{X}_A - \bar{X}_B}{S_p} \quad \text{'Observed' effect size}$$

0.20 = small effect

0.50 = moderate effect

0.80 = large effect

Example: $\mu_M = 8$; $\mu_F = 8.5$; $\sigma_M = \sigma_F = 1.5 \Rightarrow \delta = 0.33$

$\bar{X}_M$	$\bar{X}_F$	s_M	s_F	p (2-sided)	g
8.10	9.34	1.55	1.55	0.015 (*)	0.80

$$n_M = n_F = 20$$

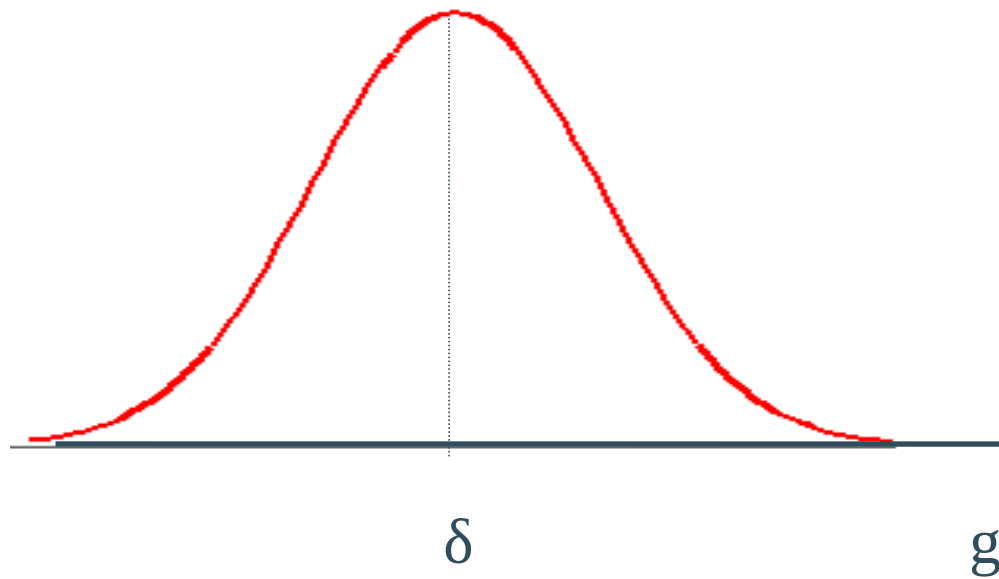
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$\bar{X}_M$	$\bar{X}_F$	s_M	s_F	p (2-sided)	g
8.10	9.34	1.55	1.55	0.015	0.80
7.60	7.59	1.23	1.47	(*) 0.98	-0.0069

Example: $\mu_M = 8$; $\mu_F = 8.5$; $\sigma_M = \sigma_F = 1.5 \Rightarrow \delta = 0.33$

X_M	X_F	s_M	s_F	p (2-sided)	g
8.10	9.34	1.55	1.55	0.015	0.80
7.60	7.59	1.23	1.47	(*)	-0.0069
7.96	8.81	1.38	1.59	0.98	0.57
7.70	8.25	1.49	1.65	0.078	0.35
8.17	8.25	1.76	1.33	0.28	0.053
7.86	8.81	1.24	1.58	0.87	0.67
8.19	7.93	1.79	1.78	0.040	-0.14
8.11	8.15	1.76	1.97	(*)	0.020
7.86	7.94	1.89	1.64	0.65	0.042
8.34	8.53	1.39	1.79	0.95	0.12

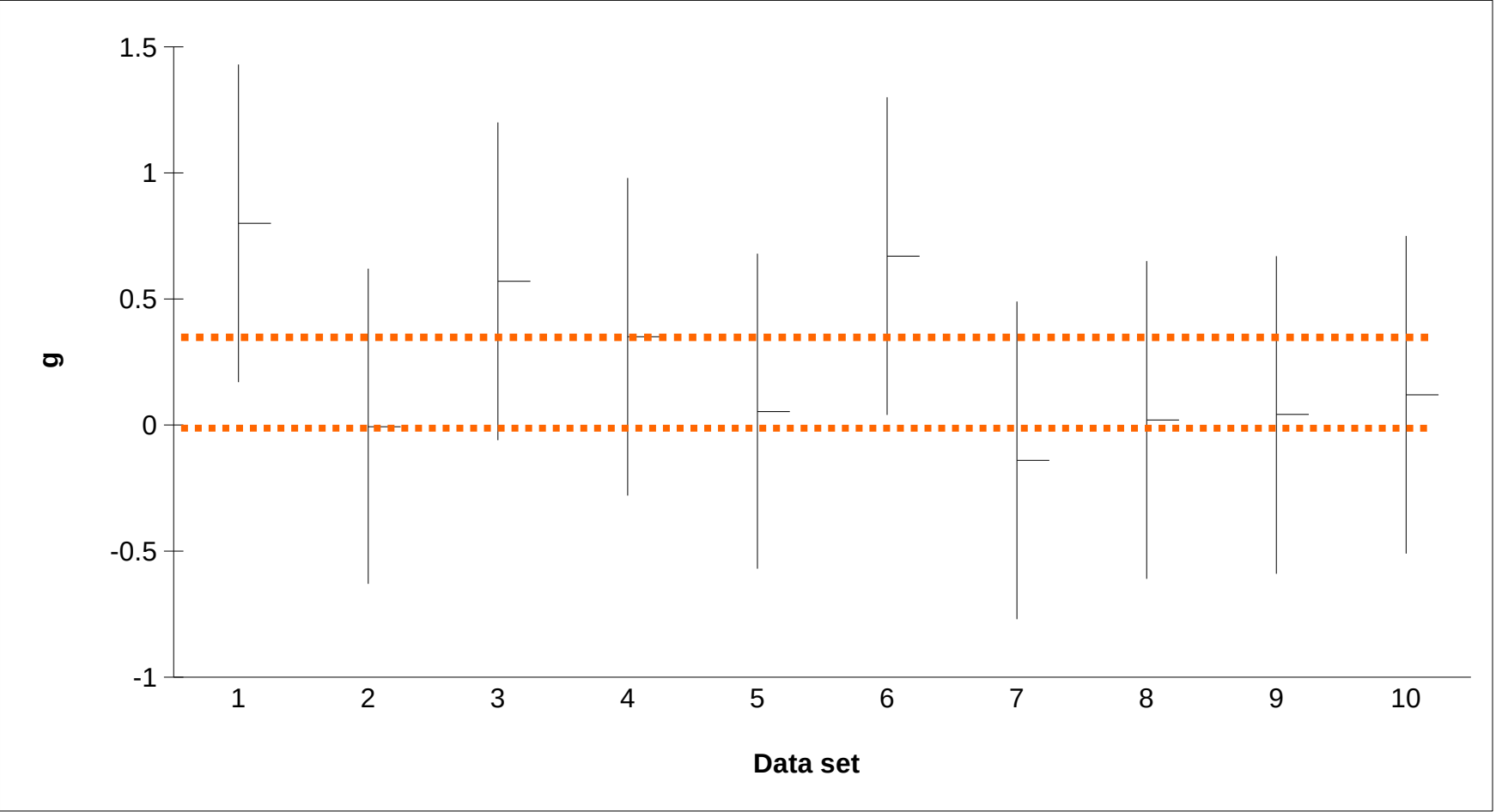
$$g \sim N(\delta, \sigma_{(g)}^2) \quad \text{with} \quad \sigma_{(g)}^2 \cong \frac{4}{N}$$



$$g \sim N(\delta, 0.1)$$

$\Rightarrow$ 95 % confidence interval:

$$\left[g - 1.96\sqrt{0.1}; g + 1.96\sqrt{0.1} \right]$$

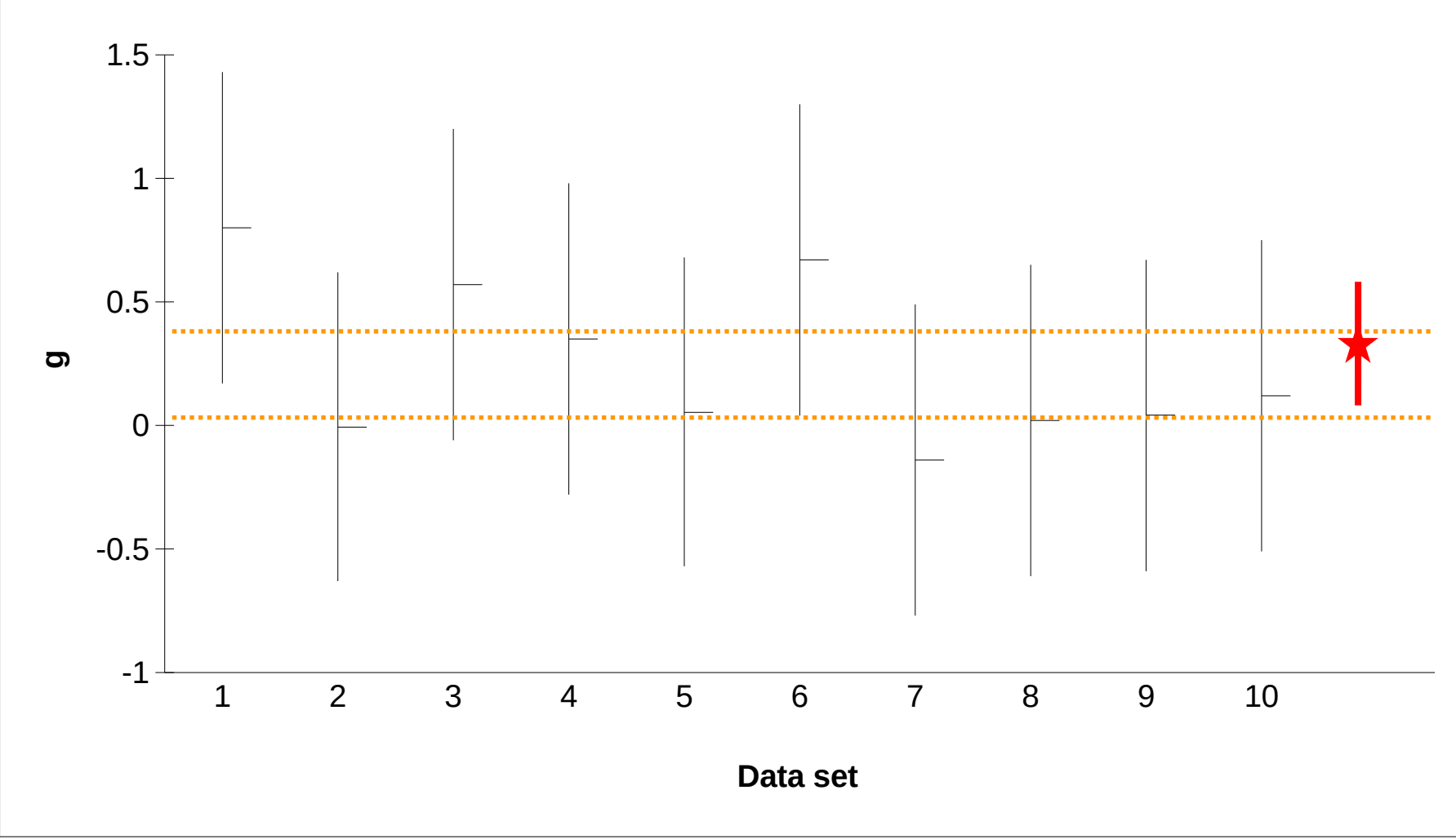


Combining study results in a meta-analysis

Suppose simulated data are data from 10 studies, being replications of each other:

$$\rightarrow \bar{g} = \frac{\sum g}{k} = 0.25$$

$$\bar{g} \sim N\left(\delta, \frac{\sigma_{(g)}^2}{k}\right) \qquad \sigma_{(\bar{g})}^2 = 0.01$$



Comparing individual study results and combined study results

1. Observed effect sizes may be positive, negative, small, moderate and large.
2. CI relatively large
3. 0 often included in confidence intervals
4. Combined effect size close to population effect size (averaging out the **noise**)
5. CI relatively small (higher **accuracy**)
6. 0 not included in confidence interval (higher **power**)

- [Simulation MA.xlsx](#)
- Sheet 'Raw data': data from 10 studies are randomly drawn from normal distributions. Note: clicking on F9 replaces the values by new random values.
- Sheet 'Study 1': gives the means, standard deviations, sample sizes, SMD, t- & p -value for first study.
- Sheet 'k studies': gives the means, standard deviations, sample sizes, SMD, t- & p -value for all 10 studies
- Sheet 'Meta-analysis': calculates the mean SMD and makes plot with confidence intervals for the individual and mean SMD. Hint: click a few times on F9.

Meta-analysis: Gene Glass (Educational
Researcher, 1976, p.3):

*“Meta-analysis refers to the analysis of
analyses”*

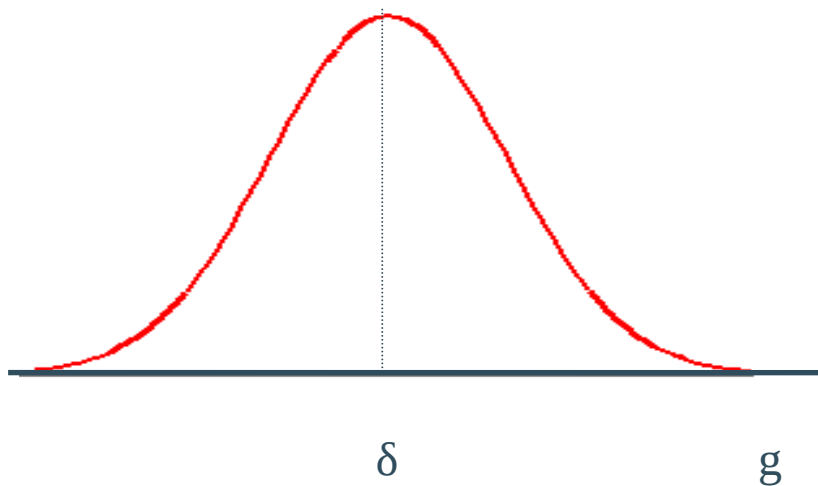
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- Meta-analytic questions and steps

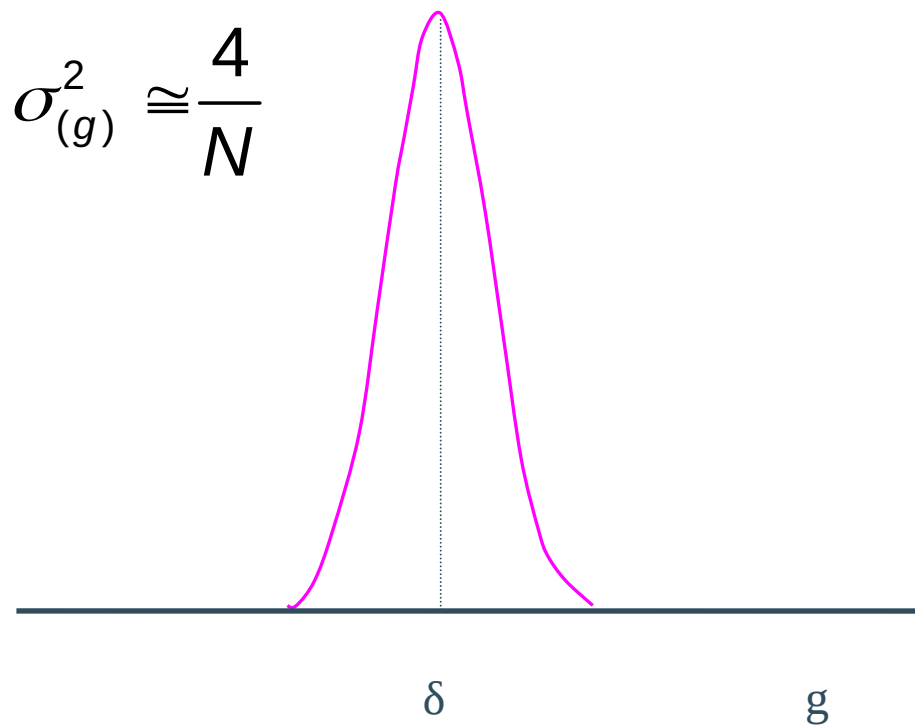
More complex meta-analyses

1. Dissimilar study sample sizes

$$g \sim N(\delta, \sigma_{(g)}^2) \quad \text{with} \quad \sigma_{(g)}^2 \cong \frac{4}{N}$$



$$n_M = n_F = 20$$



$$n_M = n_F = 100$$

1. Dissimilar study sample sizes

$$\hat{\delta} = \frac{\sum g_j}{k}$$

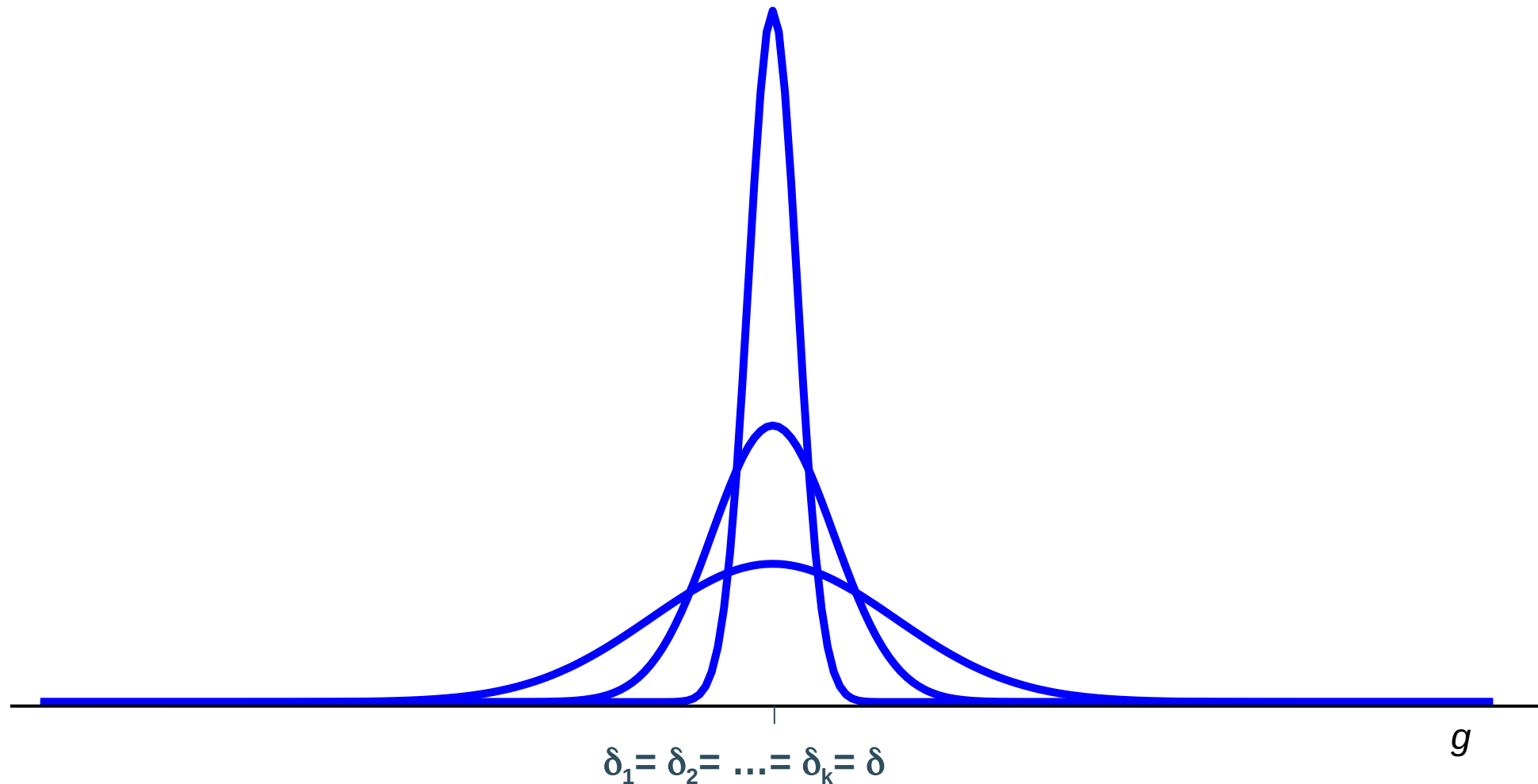
$$= \frac{\sum w_j g_j}{\sum w_j}$$

with $w_j = N_j$

or $w_j = \frac{1}{\sigma_{(g_j)}^2}$

More complex meta-analyses

2. Dissimilar study effect sizes

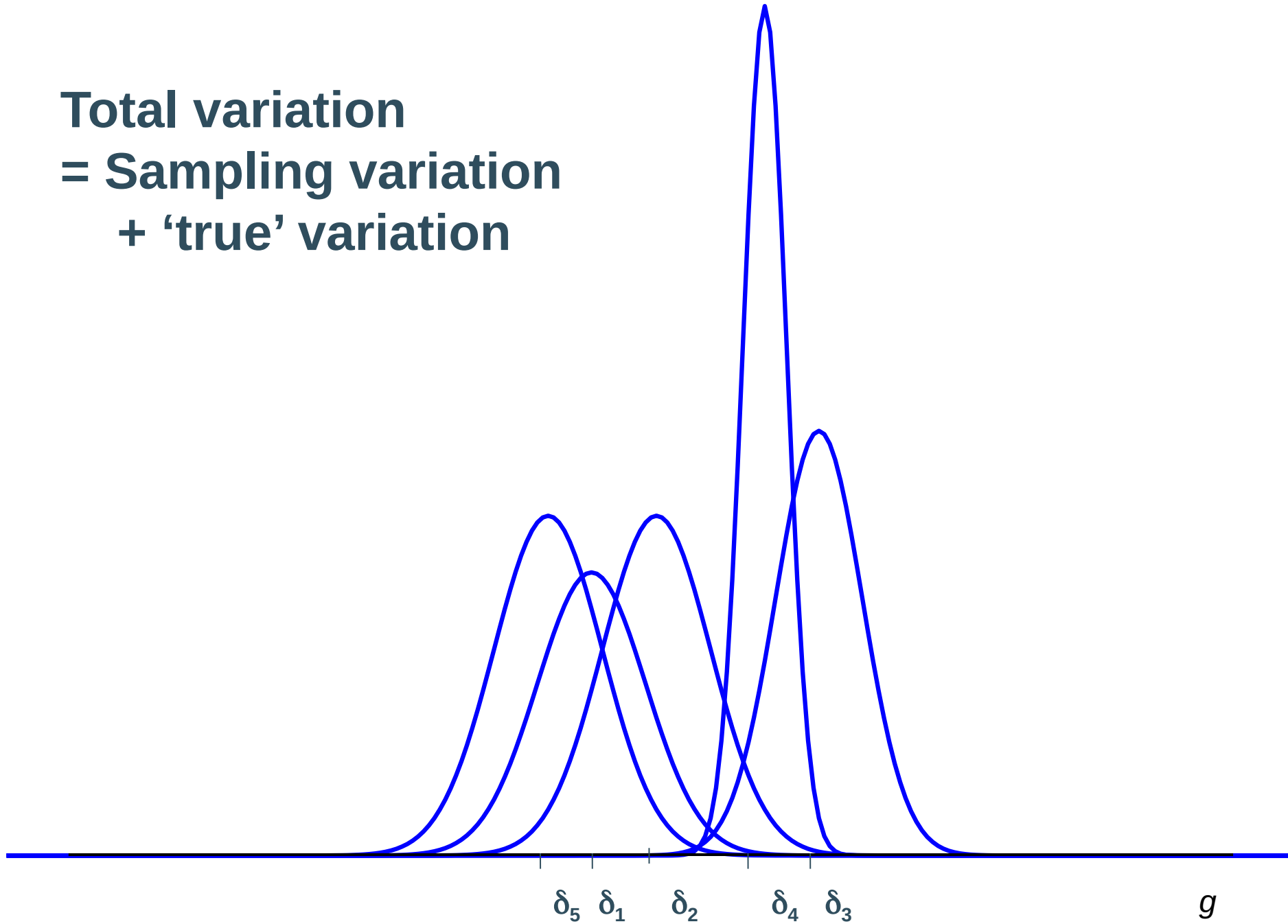


But ...

... studies often differ in:

- Kind of participants (e.g., age)
- Study context
- Outcome variable and its operationalization
- Independent variable and its operationalization
- Study design
- ...

**Total variation
= Sampling variation
+ 'true' variation**



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Basic meta-analytic questions

1. Is there an overall effect?
2. How large is this effect?
3. Is the effect the same in all studies?
4. How large is the variation over studies?
5. Is this variation related to study characteristics?
6. What is the effect in the specific studies?

Steps in a meta-analysis

1. Formulating research questions
2. Systematic search for studies
3. Selection of relevant studies
4. Extracting relevant data (outcomes, study characteristics, study quality)
5. Converting data to a common effect size measure
6. Combining and comparing the effect sizes
7. Looking for moderator variables
8. Concluding and reporting

Outro EN

Outro based on KU Leuven visual identity –
English

No presentation allowed

The logo consists of a dark blue rectangle with the text 'KU LEUVEN' in white, bold, sans-serif capital letters. This rectangle is centered within a larger, light blue rectangular area.

KU LEUVEN

Ontdek jezelf.
Begin bij de wereld.